

TESTING OF HYPOTHESIS

HYPOTHESIS

MEANING

Meaning of Hypothesis : it is a tentative proposition formulated for empirical testing. It is tentative, because its veracity (geneunity) can be evaluated only after it has been tested empirically.

A hypothesis is a tentative assumption drawn from knowledge and theory which is used as a guide in the investigation of facts and theories that are yet unknown.

Hypothesis is usually considered as the principal instrument in research.

TYPES OF HYPOTHESES

Descriptive hypotheses: These are propositions that describe the characteristics (such as size, form or distribution) of a variable. The variable may be an object, person, organization, situation or event.

For example "The rate of unemployment among arts graduates is higher than that of commerce graduates."

Relational hypotheses: These are propositions, which describe the relationship between two variables. The relationship suggested may be positive or negative correlation or causal relationship.
For example : "Families with higher incomes spend more for recreation."

Causal hypotheses: It's a change in, one variable **causes or leads** to an effect on another variable. *For example: Dells' advertising budget & promotional strategies, so on and so forth. Have boosted the sales of Dell PCs'. In this **advertising budget & promotional strategies are the causes** for increase in **SALE** (effect)*

Working hypotheses: While planning the study of a problem, hypotheses are formed. Initially they may not be very specific. In such cases, they are referred to as working hypotheses which are subject to modification as the investigation proceeds.

Research hypothesis: When a prediction or a hypothesized relationship is to be tested by scientific methods, it is termed as research hypothesis.

Directional hypothesis: The hypotheses which stipulate the direction of the **expected differences** or relationships - is termed as directional hypotheses. **For example, Adolescent boys with high IQ will exhibit low anxiety than adolescent boys with low IQ"** is a directional research hypothesis because it stipulates the direction of the difference between groups.

Non-directional hypothesis: A research hypothesis which does not specify the direction of expected differences or relationships is a non-directional research hypothesis. **For example, "There is a**

difference in the anxiety level of adolescent girls of high IQ and low IQ” are non-directional research hypotheses.

Declarative hypothesis: When the researcher makes a positive statement about the outcome of the study, the hypothesis takes the declarative form.

CHARACTERISTICS OF A GOOD HYPOTHESIS

1. Conceptual clarity: A hypothesis should be conceptually clear. If the hypothesis is not clear and precise, the inferences drawn on its basis cannot be taken as reliable.

2. Specificity: A hypothesis should be specific and explain the expected relations between variables and the conditions under which these relations will hold.

3. Testability: A hypothesis should be testable and should not be a moral judgment. It should be possible to collect empirical evidences to test the hypothesis. *The hypotheses selected should be amenable to testing within a reasonable time.*

4. Availability of techniques: Hypotheses should be framed such that it is related to available techniques. Otherwise they will not be researchable.

5. Theoretical relevance: A hypothesis should be related to a body of theory.

6. Consistency: Hypotheses should be logically consistent. That is, two or more propositions logically derived from the same theory must not be mutually contradictory.

7. Objectivity: Scientific hypotheses should be free from value –judgment(good or bad).

8. Simplicity: A hypothesis should be a simple requiring fewer conditions or assumptions.

BASIC CONCEPT CONCERNING TESTING OF HYPOTHESIS

THE LOGIC OF HYPOTHESIS TESTING

a) Null hypothesis and Alternative hypothesis formulation

In the context of statistical analysis, we often start with null hypothesis and then frame alternative hypothesis.

Suppose If we are to compare method A with method B about its superiority and if we proceed on the assumption that both methods are equally good, then this assumption is termed as the **null hypothesis**.

As against this, we may think that the method A is superior or the method B is inferior, we are then stating what is termed as **alternative hypothesis**.

The null hypothesis is generally symbolized as H_0 and the alternative hypothesis as H_A .

Analysts usually test to determine whether there has been no change in the population of interest or whether a real difference exists.

b) The level of significance: This is very important concept in the context of hypothesis testing. It is always some percentage (usually 5% i.e., 0.05) which is the most level chosen. Other levels are: 0.10, 0.025, or 0.001 are sometimes chosen. The **exact level to choose largely determined by how much risk one is willing to accept.**

c) Decision rule or test of hypothesis: Given a hypothesis H_0 and an alternative hypothesis H_A we make a rule which is known as decision rule according to which **we accept H_0 (i.e. reject H_A) or reject H_0 (ie. accept H_A).**

d) Error in testing of hypothesis: TYPE I AND TYPE II ERRORS

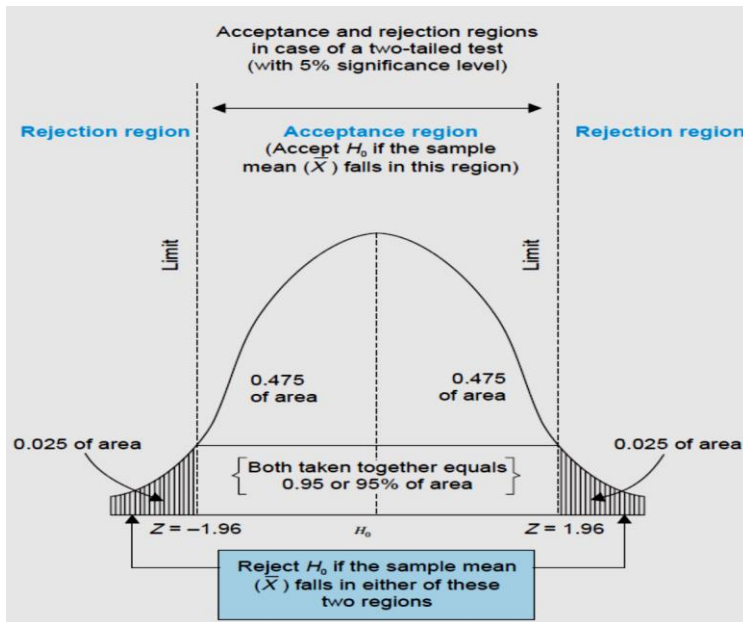
In the context of testing of hypotheses, there are basically **two types of errors we can make.**

We may reject H_0 when H_0 is true this is Type I error and is denoted by (α) known as (α) error, and We may accept H_0 when in fact H_0 is not true which is Type II error. Type II error is denoted by β (beta) known as β error.

	Decision	
	Accept H_0	Reject H_0
H_0 (true)	Correct decision	Type I error (α error)
H_0 (false)	Type II error (β error)	Correct decision

e) TWO-TAILED AND ONE-TAILED TESTS

In the context of hypothesis testing, **Two-tailed and One-tailed tests** are important and **must be clearly understood.**



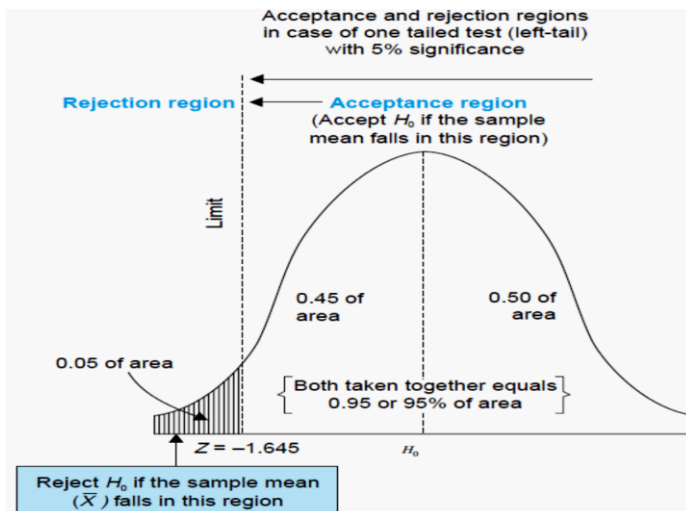
A **two-tailed test rejects** the **null hypothesis** if the **sample mean** is significantly **higher or lower** than the **hypothesized value of the mean** of the population.

Such a test is appropriate when the **null hypothesis is having some specified value** and **the alternative hypothesis is a value not equal to the specified value** of the null hypothesis.

Symbolically, the two tailed test is appropriate when we have

$H_0: \mu = \mu_{H_0}$ and $H_a: \mu \neq \mu_{H_0}$ which may mean $\mu > \mu_{H_0}$ or $\mu < \mu_{H_0}$

Thus there are two rejection regions, one on each tail of the curve which can be illustrated as under at significance level kept at 5%.



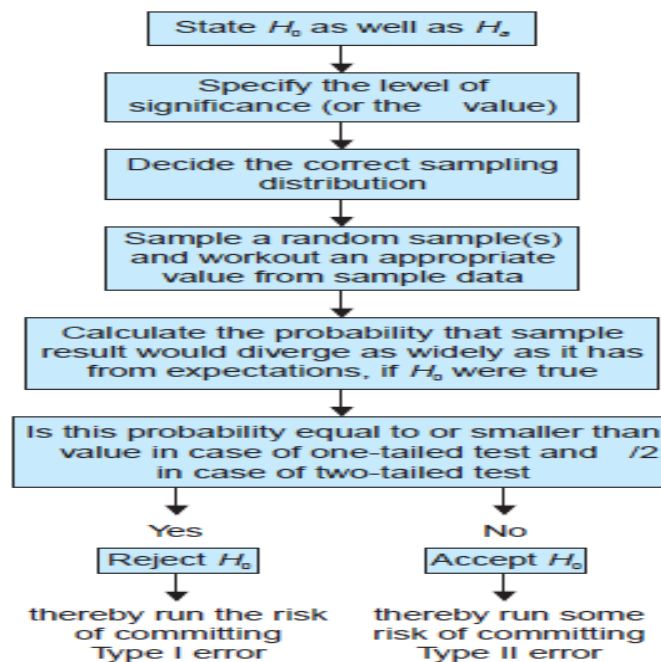
But there are situations **when only one-tailed test is considered appropriate**. A one-tailed test would be used when we test, whether the population mean is either lower than or higher than some hypothesized value.

For instance, when we have $H_0: \mu = \mu_{H0}$ and $\mu < \mu_{H0}$, here we are interested in left-tailed test (wherein there is one rejection region only on the left tail) which is illustrated at significance level kept at 5%.

Similarly when we have $H_0: \mu = \mu_{H0}$ and $\mu > \mu_{H0}$, here we are interested in right-tailed test (wherein there is one rejection region only on the right tail)

PROCEDURE FOR HYPOTHESIS TESTING

FLOW DIAGRAM FOR HYPOTHESIS TESTING



The various steps involved in hypothesis testing are stated below:

(i) Making a formal statement: This step consists in making a formal statement of the null hypothesis (H_0) and also of the alternative hypothesis (H_a). This means that hypotheses should be clearly stated, considering the nature of the research problem.

For instance, Civil Engineering Department wants to test the load bearing capacity of an old bridge which must be more than 10 tons, in that case he can state his hypotheses as under:

Null hypothesis $H_0: \mu = 10$ tons Alternative Hypothesis $H_a: \mu > 10$ tons

(ii) Selecting a significance level: The hypotheses are tested on a pre-determined level of significance and as such the same should be specified. Generally, in practice, either 5% level or 1% level is adopted for the purpose.

(iii) Deciding the distribution to use: After deciding the level of significance, the next step in hypothesis testing is to determine the appropriate sampling distribution. The choice generally remains between normal distribution and the t-distribution.

(iv) Selecting a random sample and computing an appropriate value: Another step is to select a random sample(s) and compute an appropriate value from the sample data concerning the test statistic utilizing the relevant distribution. In other words, draw a sample to furnish empirical data.

(v) Calculation of the probability: calculate the probability of the sample.

(vi) Comparing the probability: Yet another step is comparing the probability, which is calculated with the specified value for alpha α , with the significance level.

If the calculated probability is equal to or smaller than the alpha α value in case of one-tailed test (and $\alpha/2$ in case of two-tailed test), then reject the null hypothesis (i.e., accept the alternative hypothesis), but if the calculated probability is greater, then accept the null hypothesis.

PARAMETRIC AND NON- PARAMETRIC TEST

PARAMETRIC STATISTICAL TEST

The population mean (μ), standard deviation (s) and proportion (p) are called the parameters of a population distribution.

If the information about the population is completely known through its parameters then statistical test is called parametric test

Tests of hypotheses concerning the mean and proportion are based on the assumption that the population from where the sample is drawn is normally distributed.

Eg: t test, F test, z test, ANOVA

Parametric tests are more powerful because their data are derived from interval and ratio measurements.

Ex: Interval $\rightarrow$ the number of times an individual makes a complaint (occurrence)

Ex: ratio $\rightarrow$ the grades of test

Assumptions for parametric tests include the following

The observations must be independent that is, the selection of any one case should not affect the chances for any other case to be included in the sample.

The observations should be drawn from **normally distributed populations**.

These populations should have **equal variances**.

The measurement scales should be at least interval so that arithmetic operations can be used with them.

The t- Test

The *t - statistics* was introduced in 1908 by William Gosset a chemist, who was working for the Guinness brewery in Dublin, Since he published his finding under a pen name 'Student', it is known as **Student's t-test**.

When the sample size is small (i.e., less than 30), the central limit theorem does not assure us to assume that the sampling distribution of a statistic such as mean $\bar{x}$, proportion $\bar{p}$, is normal.

Student's t-test is used when two conditions are fulfilled viz., the sample size is 30 or less and the population variance is not known.

While using t-test we assume that the *population from which sample has been taken is normal or approximately normal*, sample is a random sample, observations are independent, there is no measurement error

And in the case of two samples when equality of the two population means is to be tested, we assume that the population variances are equal.

For applying t-test, The *t*-distribution tables are available which give the critical values of *t* for different degrees of freedom at various levels of significance.

The table value of 't-distribution' *t* for given degrees of freedom at a certain level of significance is compared with the calculated value of *t* from the sample data,

If the calculated value of 't' is either equal to or exceeds the table value, we infer that the difference is significant, but if calculated value of *t* is less than the concerning table value of *t*, the difference is not treated as significant.

The t-test can also be used for testing the significance of the coefficients of simple and partial correlations.

The F-test

The name was coined by George W Snedecor in honor of Sir Ronald Fisher.

F-test is based on *F*-distribution and is used to compare the variance of the two-independent samples. This test is also used in the context of analysis of variance (ANOVA) for judging the significance of more than two sample means at once and the same time.

F ratio is computed in a way that the larger variance is always in the numerator. Tables have been prepared for *F* distribution that give critical values of *F* for various values of degrees of freedom for larger variance (which is numerator) as well as smaller variances (which will be in denominator).

The calculated value of *F* from the sample data is compared with the corresponding table value of *F* from *f-distribution* and if the *F Calculated Value* is equal to or exceeds the *F table Value*, then we infer that the null hypothesis cannot be accepted.

For computational purposes, a larger sample variance is placed in the numerator so that ratio is always equal to or more than one.

$$F = \frac{s_1^2}{s_2^2}; s_1 > s_2$$

The F-distribution, also called **variance ratio distribution**, is not symmetric and the F values can never be negative.

Z-TEST

Prof. R.A. Fisher developed this test. It is based on the normal distribution and is widely used for testing the **significance of several statistics**, such as mean, median, mode, coefficient of correlation and others.

It may be noted that z-test applied with **large sample(s) greater than equal to 30, when population variance is known.**

The relevant test statistic **Z**, is worked out and compared with its probable value (to be read from table showing area under normal curve) at a specified level of significance → for judging the significance of the measure.

z-test is generally used for comparing the mean of a sample to some hypothesized mean for the population of large sample, when population variance is known

z-test is also used for judging the significance of difference between means of two independent samples in case of large samples, when population variance is known.

This test is used even when binomial distribution or *t*-distribution is applicable on the presumption that such a distribution tends to approximate normal distribution as '*n*' becomes larger

NON-PARAMETRIC TESTS

Nonparametric tests are used to test hypotheses with nominal and ordinal data. No clear information about population is available

Nonparametric tests have few assumptions: Do not specify normally distributed populations or homogeneity of variance.

Nonparametric tests are the only ones usable with nominal data and ordinal data

Ex: nominal the country of origin of a product (named categories)

Ex: ordinal High Income group, middle income group, low income (ordered categories: ranked according to order)

Nonparametric tests may also be used for interval and ratio data, although they waste much of the information available.

Nonparametric tests are easy to understand and use.

CHI-SQUARE TEST

The chi-square test is an important test amongst the several tests of significance developed by statisticians.

Chi-square, symbolically written as X^2 (Pronounced as Ki-square), is a statistical measure *used in the context of sampling analysis* for *comparing a actual variance (observed) to a theoretical variance (expected)*.

It is particularly useful in tests involving nominal data

Typical are cases where persons, events, or objects are grouped in two or more nominal categories such as "yes-no," true false, male female, "favor-undecided-against" or class 'A,B,C, or D'.

Using this technique, *we test for significant differences* between the observed distribution and the expected distribution *based on the null hypothesis*.

When we use chi-square as a test for population variance, we have to work out the value of X^2 to test the null hypothesis.

Then by comparing the calculated value with the table value of X^2 for $(n - 1)$ degrees of freedom at a given level of significance, we may either accept or reject the null hypothesis.

If the calculated value of X^2 is less than the table value, the null hypothesis is accepted, but if the calculated value is equal or greater than the table value, the hypothesis is rejected.

This test is evaluated to see whether the difference between the observed frequencies and the expected frequencies *under the null hypothesis can be attributed to chance or actual population differences*

The X^2 -distribution is *not symmetrical* and all the *values are positive*.

Table value specifies critical values of X^2 for the different degrees of freedom.

Chi-square is useful in cases of one-sample analysis, two independent samples, or k independent samples.

As a non-parametric test, chi-square can be used (i) as a test of goodness of fit and (ii) as a test of independence

As a test of goodness of fit, X^2 test enables us to see how well does the assumed theoretical distribution (such as Binomial distribution, Poisson distribution or Normal distribution) fit to the observed data.

If the calculated value of X^2 is less than the table value at a certain level of significance, the fit is considered to be a good one which means that the divergence between the observed and expected frequencies is **attributable to fluctuations of sampling**.

But if the calculated value of X^2 is greater than its table value, the fit is not considered to be a good one

As a test of independence, X^2 test enables us to explain whether or not two attributes are associated.

For instance, we may be interested in knowing whether a new medicine is effective in controlling fever or not, X^2 test will help us in deciding this issue.

In such a situation, we proceed with the null hypothesis that the two attributes (viz., new medicine and control of fever) are independent which means that new medicine is not effective in controlling fever.

On this basis we first calculate the expected frequencies and then work out the value of X^2 . If the calculated value of X^2 is less than the table value at a certain level of significance for given degrees of freedom, we conclude that null hypothesis stands, which means that the two attributes are independent or not associated (i.e., the new medicine is not effective in controlling the fever).

CONDITIONS FOR THE APPLICATION OF X^2 TEST

The following conditions should be satisfied before X^2 test can be applied:

- 1) Observations recorded are **collected on a random basis**.
- (ii) All the items in the **sample must be independent**.
- (iii) **No group should contain very few items, say less than 10**. 10 is regarded as better by most of the statisticians.
- (iv) The overall number of items must also be reasonably large. It should normally be at least 50.
- (v) The constraints must be linear. (i.e., equations containing no squares or higher powers of the frequencies) are known as linear constraints.

U-TEST IS POPULARLY KNOWN AS WILCOXON-MANN-WHITNEY TEST WHEREAS H-TEST IS KNOWN AS KRUSKAL-WALLIS TEST. THEY ARE RANK-SOME TEST

(A) WILCOXON-MANN-WHITNEY TEST(or U-test): This is a very popular test amongst the rank sum tests.

The U test is designed to determine whether two independent samples have been drawn from the same population or from two different populations having the same distribution.

i) Mann - whitney U-Test:- It is a non parametric counterpart of the t-test used to compare the means of two independent population. The two tailed hypothesis for this test are as follows.
Ho: The two population are identical
H₁: The two population are not identical

THE KRUSKAL-WALLIS TEST (or H test): This test is conducted in a way similar to the U test. This test is used to test the whether null hypothesis that k independent random samples come from identical universes or testing the alternative hypothesis - wherein the samples have come from different universes which are not equal.

ii) Kruskal - Wallis test:- It is a non-parametric test which is used to compare three or more groups of sample data. The hypothesis are,
Ho: The samples are from identical population
H₁: i) The samples drawn from the population are random
ii) The case of each group are identical
iii) The measurement scale should be ordinal.

ANALYSIS OF VARIANCE (ANOVA)

ANALYSIS OF VARIANCE (ANOVA) was developed by Sir Ronald Fisher in the 1930s as a way to interpret the results from agricultural experiments.

ANOVA is a statistically based, objective decision-making tool for detecting any differences (variation) in average performance of groups of items tested.

The essence of ANOVA is that the total amount of variation in a set of data is *broken down into two types*, that amount which can be attributed to chance and that amount which can be attributed to specified causes.

The Basic principle of ANOVA is to *test for differences among the means of the populations* by examining the amount of variation within each of these samples, relative to the amount of variation between the samples.

The ANOVA is very much useful in researches in social science and other disciplines. It is appropriate to situations where more than two categories have to be compared, e.g., the fuel efficiency of several brands of cars, and the like.

NO-WAY ANOVA

It is the simplest situation to analyze and has some practical value. Here no parameters are controlled.

ONE-WAY ANOVA is the next most complex ANOVA to conduct. This situation considers the effect of one controlled parameter upon the performance of a product or process.

TWO-WAY ANOVA is the next highest order of ANOVA to review; there are two controlled parameters in this experimental situation.

THREE-WAY ANOVA. Three-way ANOVA entails three controlled factors in an experiment.